

Airbnb Price Prediction Report — Lyon

Generated on 2026-08-25 by the automated pipeline (`make all`). Source: InsideAirbnb.

Executive summary

This report analyzes the **Lyon** Airbnb market using 5,427 cleaned listings and 26 listing features. A machine-learning model was trained to predict nightly prices and then evaluated on a **holdout set** — listings it had never seen before.

The best model was **XGBoost**. On the holdout set it achieved:

- **Average error (MAE): \$30** — a typical prediction is off by about this much.
- **RMSE: \$49** — like MAE, but weighs large errors more heavily.
- **MAPE: 27%** — the average error as a share of the actual price.
- **R²: 0.70** — the share of price variation the model explains (0–1).
- **61%** of predictions were within +/-25% of the true price.

How to read this report

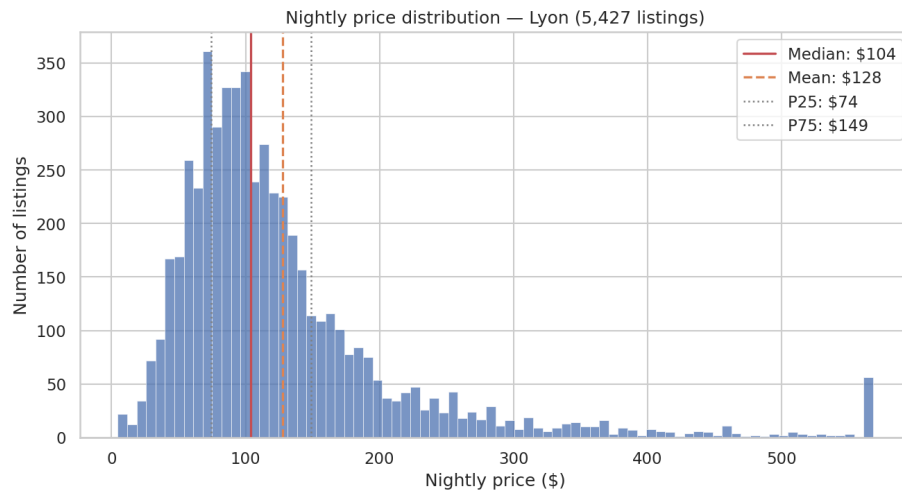
- **MAE (mean absolute error)** — the average difference between predicted and actual prices, in dollars. Lower is better; it's the easiest number to grasp ("off by about \$X on average").
- **RMSE (root mean squared error)** — similar to MAE but penalizes large mistakes more. If RMSE is much higher than MAE, a few big errors are inflating it.
- **MAPE (mean absolute percentage error)** — the same idea as MAE, but expressed as a percentage of the actual price, so it can be compared across cities.
- **R²** — how much of the variation in prices the model explains (0 = no better than guessing the average; 1 = perfect).
- **Holdout set** — the 10% of listings the model never saw while training. All accuracy numbers come from this set, so they are an honest estimate of real-world performance.
- **Top features** — the listing attributes the model leans on most when setting a price.

Data

- Source: InsideAirbnb (latest available snapshot)
- Listings after cleaning: **5,427** — median nightly price **\$104**, mean **\$128**
- Features used by the model: **26**
- Train / validation / holdout split: **3,798 / 977 / 652** listings (70/15/10, random seed 42)

Market overview

Price distribution



- Median nightly price: **\$104**; the middle 50% of listings fall between **\$74** and **\$149**. - The mean (**\$128**) sits well above the median — a minority of expensive listings pull the average up, so the **median is the better guide** to a typical price.

Price by room type

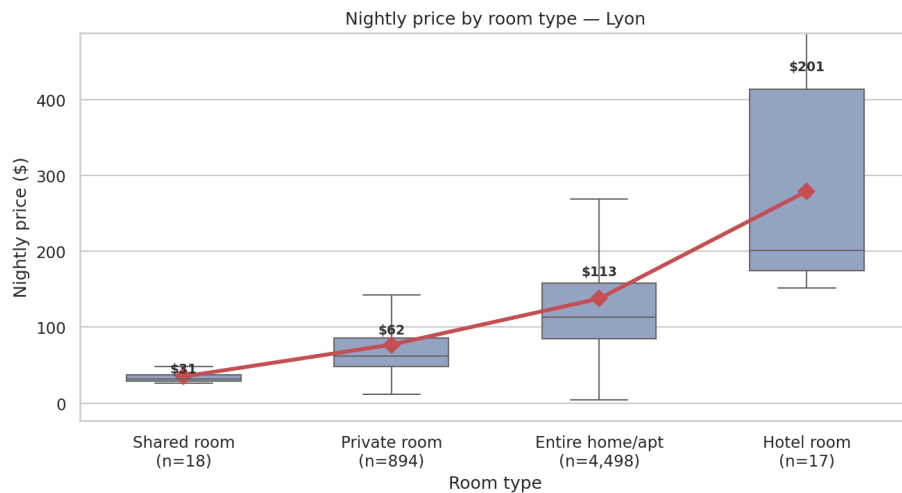


Figure 1: price_by_room_type

Room type	Listings	Median price	Mean price
Hotel room	17	\$201	\$279
Entire home/apt	4,498	\$113	\$138
Private room	894	\$62	\$77
Shared room	18	\$31	\$35

Price by number of bedrooms

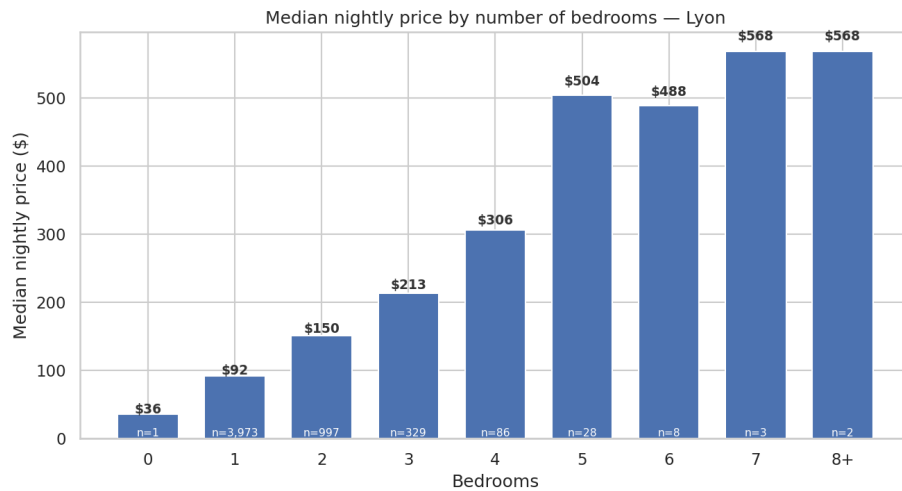


Figure 2: price_by_bedrooms

Feature correlations

Model accuracy on the holdout set

Metrics for the chosen model (**XGBoost**) on 652 unseen listings:

Metric	Value	What it means
MAE	\$30	Average prediction is off by this much
RMSE	\$49	Errors, with big misses weighted more
MAPE	27%	Average error as a % of the true price
R ²	0.70	Share of price variation explained

Accuracy in practice: 27% of predictions were within +/-10% of the true price, 61% within +/-25%, and 87% within +/-50%.

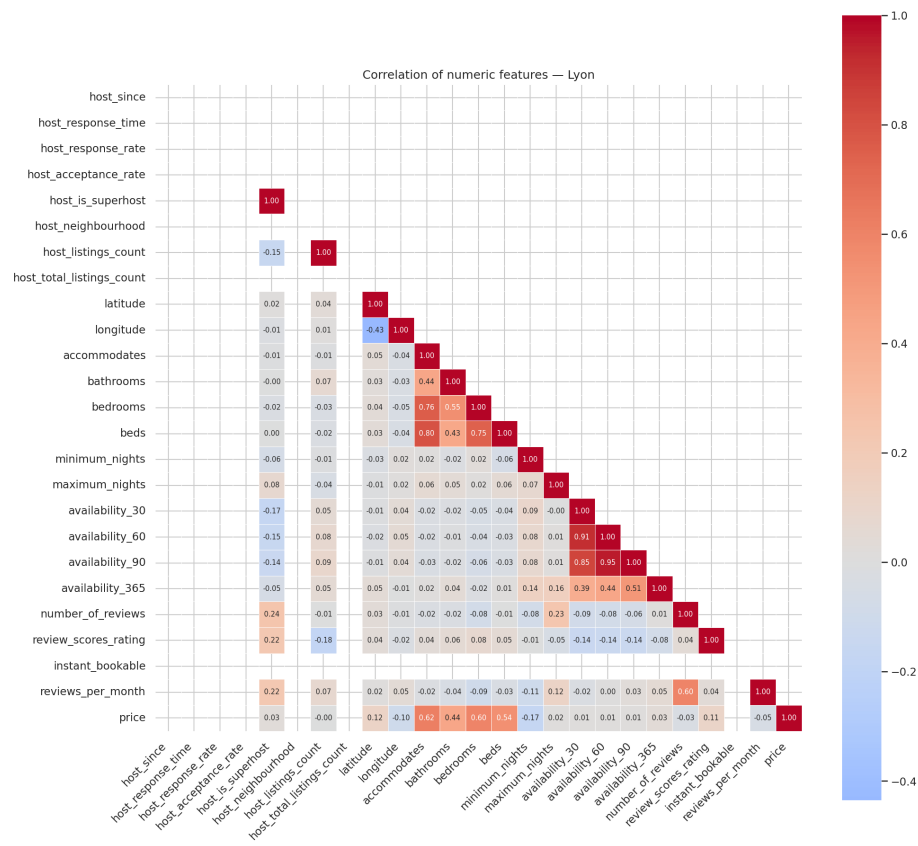
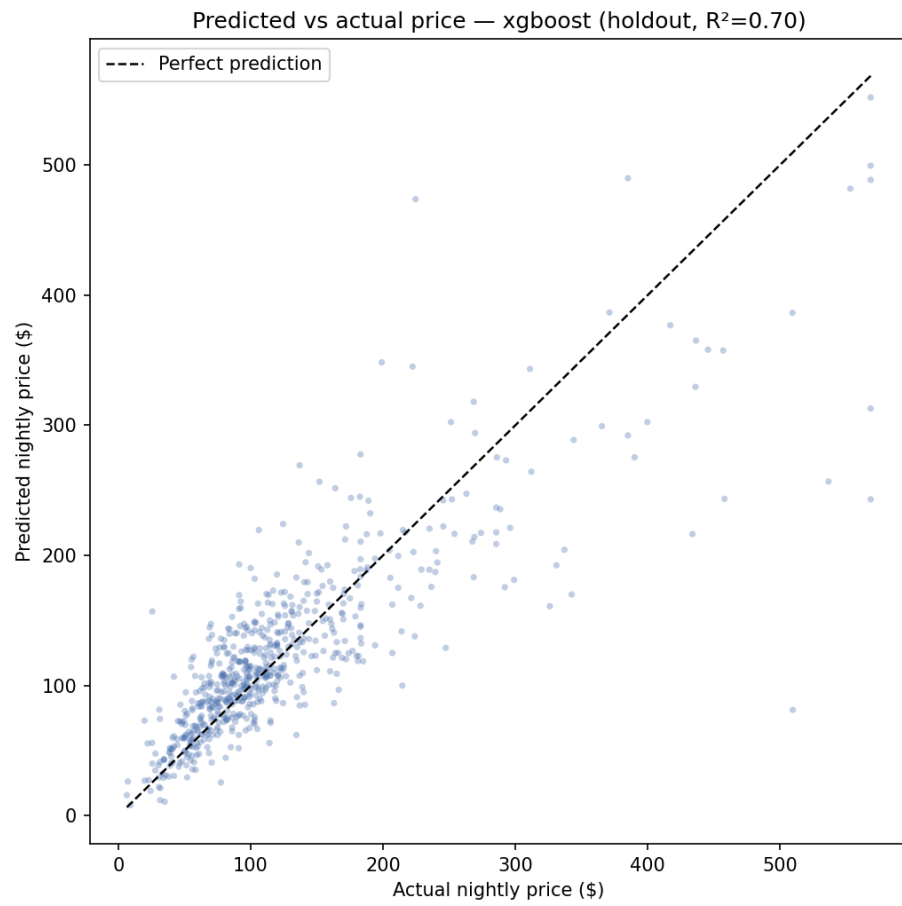
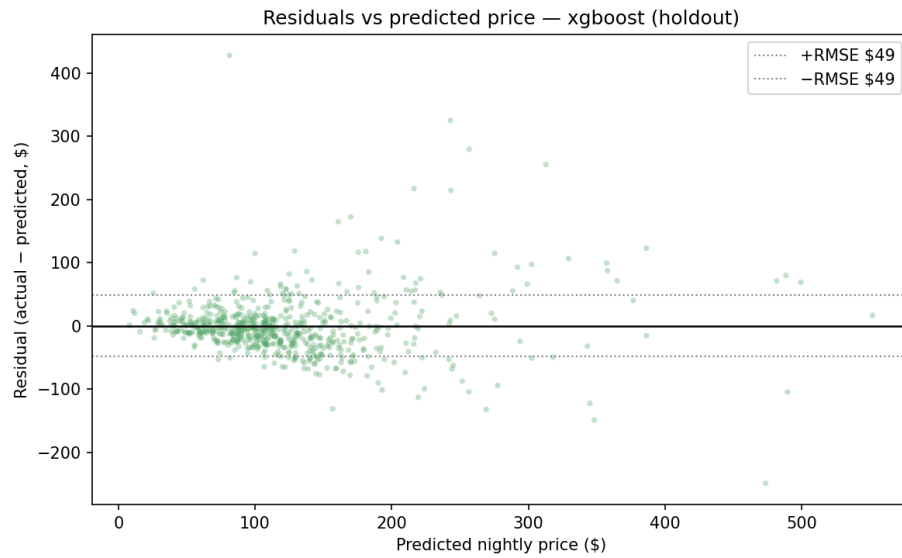


Figure 3: corr_heatmap



Points close to the dashed line are accurate predictions; points above it mean the model under-predicted.



Residual = actual - predicted. A cloud centered on the zero line means errors are balanced rather than systematically too high or too low.

Models & results

Model	Validation RMSE
Random Forest	\$57
XGBoost	\$52
TabPFN (capped sample, not directly comparable)	\$49

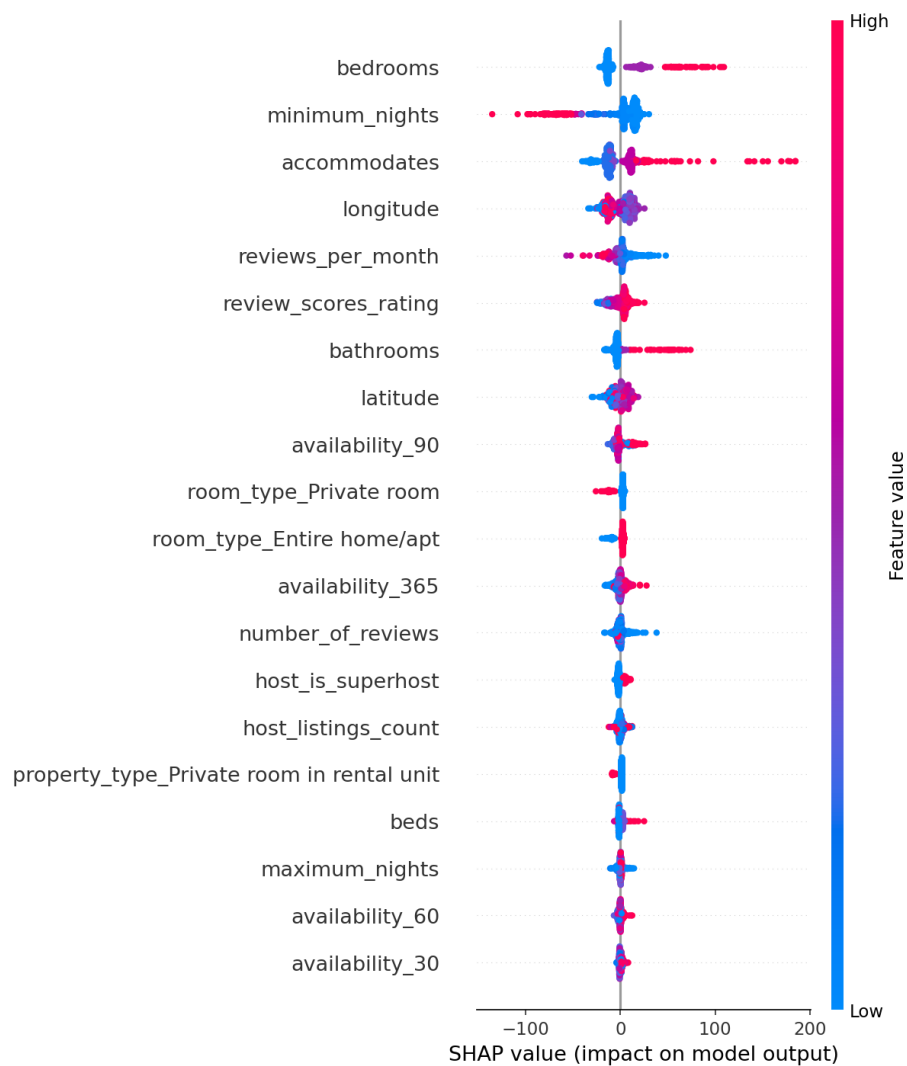
The model with the lowest validation RMSE was then scored once on the holdout set — see the accuracy section above for its final numbers.

Top features

The listing attributes that influence price the most (importance = how much the model relies on them):

Feature	Importance	Meaning
bedrooms	0.180	the number of bedrooms
bathrooms	0.090	the number of bathrooms
accommodates	0.086	how many guests a listing sleeps

Feature	Importance	Meaning
room_type_Private room	0.071	the room type (entire home vs private/shared room) (specifically Private room)
minimum_nights	0.059	the minimum nights required
property_type_Private room in loft	0.057	the property type (apartment, house, etc.) (specifically Private room in loft)
property_type_Private room in rental unit	0.057	the property type (apartment, house, etc.) (specifically Private room in rental unit)
room_type_Hotel room	0.049	the room type (entire home vs private/shared room) (specifically Hotel room)
property_type_Room in hotel	0.031	the property type (apartment, house, etc.) (specifically Room in hotel)
room_type_Entire home/apt	0.026	the room type (entire home vs private/shared room) (specifically Entire home/apt)



SHAP summary: each dot is one listing; colour shows the feature's value (red = high, blue = low) and position shows whether it pushed the predicted price up (right) or down (left).

Prediction examples

A few real holdout listings with the model's prediction:

Listing	Actual	Predicted	Error	Error %
accommodates= \$ 50 bedrooms=1, room_type=Entire home/apt	\$50	\$50	\$0	0.3%
accommodates= \$ 222 bedrooms=2, room_type=Entire home/apt	\$222	\$345	\$123	55.2%
accommodates= \$ 136 bedrooms=2, room_type=Entire home/apt	\$136	\$210	\$74	54.0%
accommodates= \$ 112 bedrooms=1, room_type=Entire home/apt	\$112	\$117	\$6	5.2%
accommodates= \$ 170 bedrooms=3, room_type=Entire home/apt	\$170	\$155	\$14	8.5%
accommodates= \$ 88 bedrooms=1, room_type=Entire home/apt	\$88	\$101	\$13	14.6%
accommodates= \$ 326 bedrooms=2, room_type=Entire home/apt	\$326	\$161	\$165	50.6%
accommodates= \$ 124 bedrooms=1, room_type=Entire home/apt	\$124	\$112	\$12	9.6%

Recommendations

- The best-performing model was **XGBoost** (holdout RMSE \$49, MAE \$30, R^2 0.70).
- On validation, **XGBoost** beat Random Forest by 8% on RMSE (\$52 vs \$57). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **bedrooms**, **bathrooms**, **accommodates** — i.e. bedrooms (the number of bedrooms); bathrooms (the number

of bathrooms); accommodates (how many guests a listing sleeps). Hosts can use these levers when setting nightly rates.

- On average, predictions are off by about **27%** of the actual price (median error 20%), and **61%** of predictions landed within $\pm 25\%$ of the true price.
- Accuracy caveat: the holdout RMSE of \$49 is **47% of the median listing price (\$104)**, so predictions for typical listings are expected to be off by roughly $\pm \$49$.
- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

Limitations

- The model only “sees” the features listed in the config — it cannot capture neighbourhood charm, amenities, photos, or seasonality.
- Extreme high-end prices are clipped at the 99th percentile during cleaning, so predictions at the very top of the market are less meaningful.
- The accuracy numbers are averages: individual predictions can be much better or much worse.
- This report is generated automatically from the pipeline artifacts and reflects the data as of 2026-08-25.

*Automated pipeline: fetch $\rightarrow$ preprocess $\rightarrow$ EDA $\rightarrow$ train $\rightarrow$ evaluate $\rightarrow$ report.
Code in `src/`, config in `configs/base.yaml`.*